Physics in Progress CARTHAGE COLLEGE **Rapid Communication**

Characterizing the relative electric permittivity of coaxial cables

Kaitlyn Prokup
Department of Physics and Astronomy, Carthage College, 2001 Alford Park Drive, Kenosha, WI 53140, USA

(Received January 15, 2024, Published online February 6, 2024)

Coaxial cables are electrical cables utilized in communications technology that transmit a radio frequency signal. These cables consist of an inner conductor, a dielectric insulator, and an outer conductor. This structure protects the signal from electromagnetic interference and power loss; however, it also inherently inhibits the speed of the signal. This inhibition is determined by the relative electric permittivity of the insulator, a unitless value that describes how easily an electric field can pass through a material. In our experiment, we connected coaxial cables of varying lengths to a signal generator and an oscilloscope to determine the propagation speed and, thus, measure the relative permittivity. Performing a linear regression on the data, we obtained a signal speed of $(216 \pm 4) \times 10^6$ m/s. Assuming a relative magnetic permeability of $\mu_r = 1$, we obtained a result of $\epsilon_r = 1.9260(2)$.

I. INTRODUCTION

Coaxial cables are a vital yet underappreciated invention in the history of communications technology. Patented by AT&T's Bell Telephone Laboratories in 1931 [1], coaxial cables were initially used for telephone infrastructure and television transmission [2]. Despite being invented nearly 100 years ago [1, 2], coaxial cables are still employed for similar purposes today. Coaxial cables are commonly used in applications such as computer network connections, internet connections, radio communications, and cable television signals [3, 4]. This begs the question, why do coaxial cables continue to see such regular use? One of the main reasons is due to their isolating properties.

Rather than an electrical current, coaxial cables transmit a radio frequency signal [4]. These cables are structured so that this signal passes through an inner conductor, surrounded by a dielectric insulator, an outer conductor, and a protective plastic jacket [2, 3, 5]. This relative isolation of the signal means that coaxial cables are protected from electromagnetic interference and will not incur power loss when placed near metal materials. However, this structure inherently inhibits the speed of the signal. Within a coaxial cable, signals are transmitted in the form of an electromagnetic wave in the transverse electromagnetic mode (TEM). In this mode, both the electric and magnetic fields are transverse to the direction of propagation. Due to the coaxial cable's structure, the TEM's vibrations travel at roughly the speed of light of the insulator. This speed is determined by the relative electric permittivity of the insulator, a unitless physical characteristic that describes how easily an electric field can pass through a material. The relative permittivity also describes other vital characteristics of the cable, such as the shunt capacitance.

As such, in our experiment, we connected RG-6 coaxial cables of varying lengths to a signal generator and an oscilloscope in order to measure the signal delay between cables relative to the cable length. We use the results of this process to calculate a value for the speed of the signal in the coaxial cable, allowing us to determine the relative permittivity of the cable.

II. EXPERIMENTAL FINDINGS

A. Methodology

In our experiment, we have measured the delay in signal between a cable of length 5 ft to that of cable varying from 40 ft to 160 ft in length. Our process involved connecting the two coaxial cables to a signal generator and an oscilloscope, with the cables connected to the oscilloscope through two 50 Ω terminators in order to prevent signal reflection. We transmitted signals of 106 Hz through these cables and obtained the results shown in fig. 1.

B. Results of Linear Fit to Data

In fig. 1, we can identify a clear increasing linear relationship between the difference in the length of the cables and the delay in the signal. This is as we would expect, given the signal will take a greater time to travel in a longer cable as compared to a shorter one. More notably, however, the slope of this relationship is what describes the speed of the signal in the cable, as it indicates the time the signal takes to travel as compared to the length it is traveling. For our analysis, we chose to utilize the Python package SciPy [6] in order to perform a linear regression with our collected data. This yielded a linear relationship with a slope of $m = 1.41 \pm 0.02 \frac{\text{ns}}{\text{ft}}$, with an

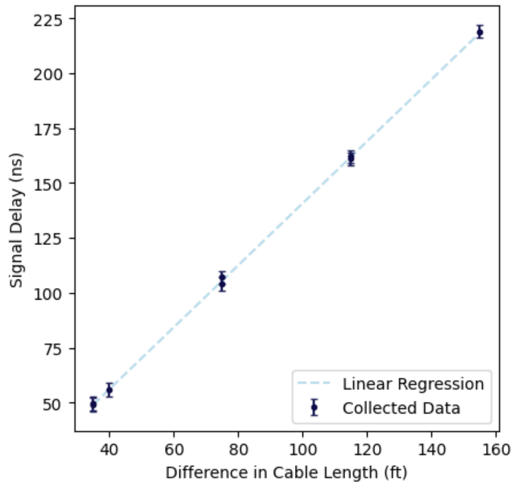

FIG. 1. Graph of the signal delay in nanoseconds plotted against the difference in length between the two cables in feet. The points plotted in a darker color indicate the collected data, with error bars of 3 ns. The dotted line plotted in a lighter color indicates the linear regression for the plotted data. This process resulted in an R-squared value of 0.99 and a P value of 3.40×10^{-12}. The resulting slope was $1.41 \pm 0.02 \frac{\text{ns}}{\text{ft}}$.

R-squared value of 0.99 and a P value of 3.40×10^{-12}. Converting the units, this result translates to a signal speed of $(216 \pm 4) \times 10^6 \frac{\text{m}}{\text{s}}$, which is just over 70% of the speed of light.

Since the signal within the cable is directed by the insulator surrounding the inner conductor, the relative electric permittivity and the relative magnetic permeability of the insulator are what determines the speed of the signal as compared to the speed of light. This relation is specifically described by eq. (1). In eq. (1), v denotes the speed of the signal, c denotes the speed of light, ϵ_r denotes the relative electric permittivity, and μ_r denotes the relative magnetic permeability. For our purposes, the relative magnetic permeability can be assumed to be equal to 1, as this is the approximate value of μ_r for most dielectric insulators used in coaxial cables. As such, utilizing the value we have calculated for the signal speed, we find a result of $\epsilon_r = 1.9260 \pm 0.0002$

$$v = \frac{c}{\sqrt{\epsilon_r \mu_r}}. \tag{1}$$

C. Discussion of Results

Coaxial cables typically employ a polyethylene based insulator [3] which have relative permittivities ranging from 1.6 to 3.0, with many types possessing a value of 2.3 [7]. As such, we find that our derived value for the relative permittivity is within a reasonable range of what might be expected from a relatively standard coaxial cable.

Deriving a value for the relative permittivity is particularly valuable as this allows us to characterize the coaxial cable. Coaxial cables can be described by only five parameters: The length of the cable, the outside diameter of the inner conductor, the inside diameter of the outer conductor, the magnetic permeability and the relative electric permittivity. By determining the relative permittivity, this gives the information necessary to fully employ the cable for circuit and network planning.

As such, a future experiment could explore the extension of these results to the characteristics of the cable. For example, determining the characteristic impedance and other parameters in order to evaluate the effectiveness of coaxial cable for various applications in electrical networks and systems. It may also be of interest to isolate the value of the relative magnetic permeability in a future experiment, in order to confirm the results of previous findings that μ_r is approximately equal to 1.

III. CONCLUSION

In conclusion, we connected coaxial cables of varying lengths to a signal generator and an oscilloscope in order to measure the signal delay between cables relative to the cable length. Describing the relationship between these values allowed us to determine the speed of propagation within the cable and thus derive the relative electric permittivity of the insulator. Performing a linear regression on the data we collected, we obtained a signal speed of $(216 \pm 4) \times 10^6 \frac{\text{m}}{\text{s}}$ with an R-squared value of 0.99 and a P value of 3.40×10^{-12}. Assuming a relative magnetic permeability of $\mu_r = 1$, we thus obtained a result of $\epsilon_r = 1.9260 \pm 0.0002$.

Deriving a value for the relative permittivity allowed us to fully characterize the coaxial cable, as coaxial cables are described by only five parameters: The length of the cable, the outside diameter of the inner conductor, the inside diameter of the outer conductor, the magnetic permeability and the relative electric permittivity. Characterizing the cable is vital, as it provides necessary information in order to fully utilize the cable in electric network and system planning. As such, future investigation could explore the extension of these results to the characteristics of the cable and evaluate the effectiveness of coaxial cable for various applications in electrical networks and systems.

ACKNOWLEDGMENTS

The author would like to thank Alyssa Gulsvig, Dalton Callow, Hector Rauda, and Thomas Howard for their contributions towards the data collection and analysis process.

Appendix A: Uncertainty Estimations

a. Coaxial Cable Length The uncertainty in the length of the coaxial cables was estimated according to the size of the connectors of the coaxial cables, giving a uniform value of 0.16 ft ft for each length data point.

b. Signal Delay The uncertainty in the signal delay was estimated by taking multiple points of data for each difference in cable length and determining the observed variance in the resulting data. To allow more generous room for uncertainty, we determined the largest variance and used this as a uniform value for the uncertainty in the signal variance measurements. This value culminated to be 3.0 ns.

Appendix B: Error Calculations

a. Slope Error Given our uniform uncertainties, we used eq. (B1) to determine the uncertainty in the slope resulting from the linear regression, where $\sigma_{\Delta t}$ is the uncertainty in the signal delay, L is the difference in the cable lengths, $\overline{L}$ is the mean of L, $\sqrt{\langle (L-\overline{L})^2\rangle}$ is the statistical standard deviation of the L values, and N is the population count of the data. This resulted in a value of $0.02\,\frac{\mathrm{ns}}{\mathrm{ft}}$.

$$\delta_m = \frac{\sigma_{\Delta t}}{\sqrt{\langle (L-\overline{L})^2\rangle}\sqrt{N}} \tag{B1}$$

b. Relative Electric Permittivity Error We derived that the equation for our uncertainty in the relative permittivity was eq. (B2), where c is the speed of light, σ_v is the uncertainty in the signal propagation speed, μ_r is the relative magnetic permeability, and v is the signal propagation speed. For the values we had derived in our analysis, this yielded a result of 0.0002.

$$2\sqrt{\frac{c^4\sigma_v^2}{\mu_r^2 v^6}} \tag{B2}$$

[1] E. Lloyd and H. A. Affel, Concentric conducting system (1931).

[2] K. Thresh, History of coax cables (2019).

[3] K. Thresh, What is coaxial cable and how is it used? (2023).

[4] E. Edwards, Types of coaxial cables and how they are specified.

[5] Coaxial cables (2023).

[6] P. Virtanen, R. Gommers, T. E. Oliphant, M. Haberland, T. Reddy, D. Cournapeau, E. Burovski, P. Peterson, W. Weckesser, J. Bright, S. J. van der Walt, M. Brett, J. Wilson, K. J. Millman, N. Mayorov, A. R. J. Nelson, E. Jones, R. Kern, E. Larson, C. J. Carey, İ. Polat, Y. Feng, E. W. Moore, J. VanderPlas, D. Laxalde, J. Perktold, R. Cimrman, I. Henriksen, E. A. Quintero, C. R. Harris, A. M. Archibald, A. H. Ribeiro, F. Pedregosa, P. van Mulbregt, and SciPy 1.0 Contributors, Nature Methods **17**, 261 (2020).

[7] T. Zedníček, What is a Dielectric Constant of Plastic Materials? (2023).